

Algebra 1
Unit 11
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

1. In the equation $y = a(b)^x$, a represents the ☐ initial amount.
☒ growth factor. and b represents ☒ initial amount
☐ growth factor
2. When b is greater than 1, the equation models ☒ exponential growth.
☐ exponential decay.
3. When b is between 0 and 1, the equation models ☐ exponential growth.
☒ exponential decay.

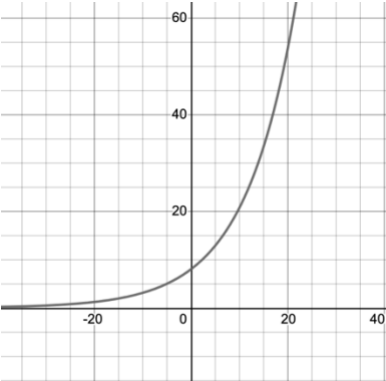
For questions 4-8, identify if each representation of a function is an example of exponential growth or exponential decay. Then justify your selection.

4.

x	y
-1	12
0	4
1	$\frac{4}{3}$
2	$\frac{4}{9}$
3	$\frac{4}{27}$

 represents exponential decay because...
multiply by factor of $\frac{1}{3}$

5. $y = 25(0.79)^x$ represents exponential decay because...
 b is 0.79 which is < 1

6.  represents exponential growth because...
the y -values increase

7. $y = 6(1 + 0.05)^x$ represents exponential growth because...
 $b > 1$

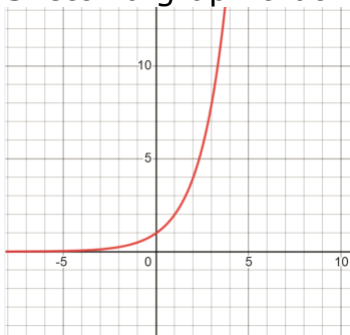
8.

x	y
2	9
3	13.5
4	20.25
5	30.375

 represents exponential growth because...

increases by rate of 1.5

9. Sketch a graph that represents exponential growth.



10. Sketch a graph that represents exponential decay.

